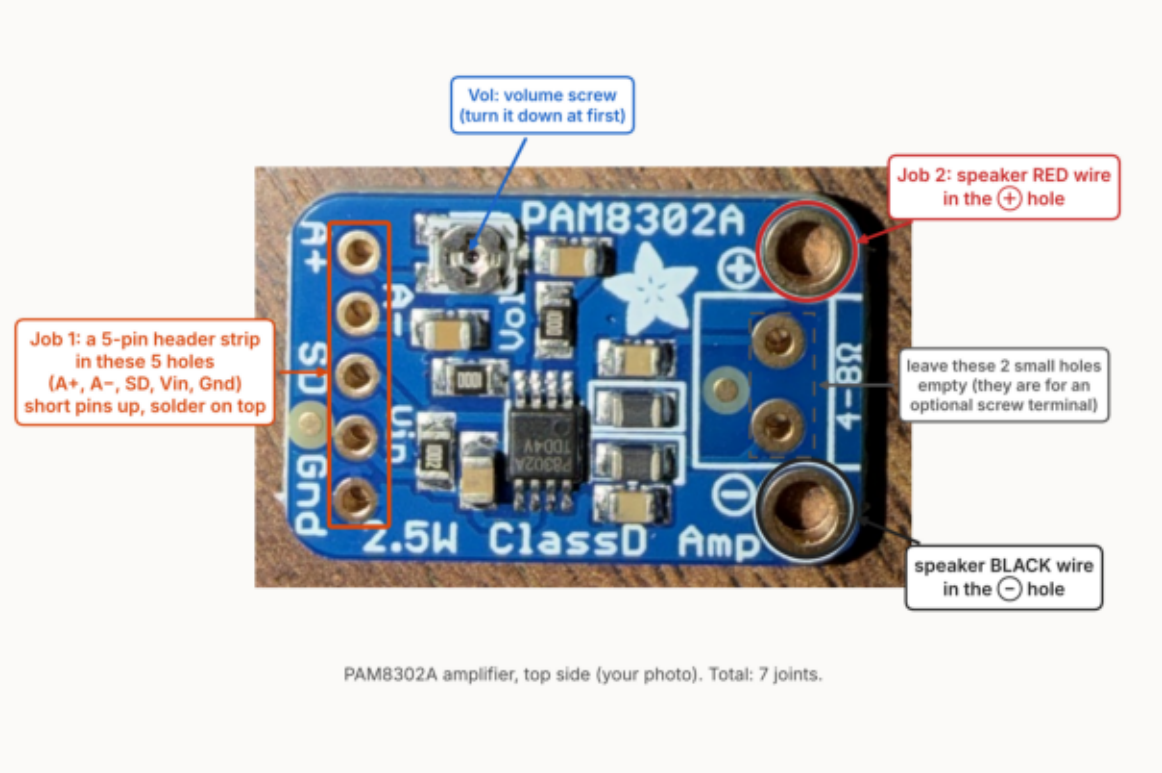
Speaker amplifier and button: what to solder

Three small jobs, about 15 minutes. Everything else is already wired and working.

Bring

The PAM8302A amplifier board, the speaker and the red button (with their wires).
A 5-pin male header strip, 2.54 mm (0.1"). One usually comes in the amplifier's bag.
Soldering iron (about 330 °C), solder, wire strippers, cutters, and the multimeter.



PAM8302A amplifier, top side (your photo). Total: 7 joints.

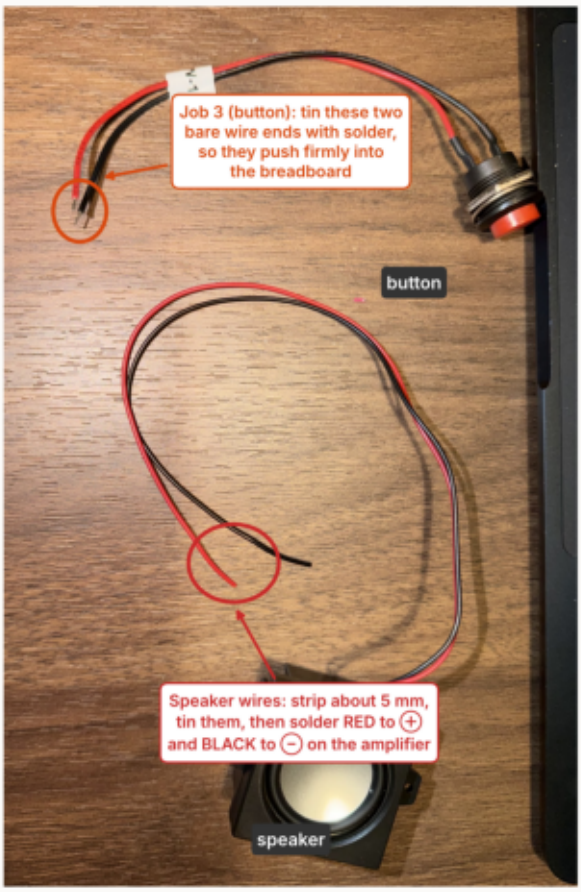
Job 1 · header pins on the amplifier (5 joints)

1. Push the long ends of the 5-pin header into a breadboard. Sit the amplifier on top, labels up, so the pins come through A+, A−, SD, Vin and Gnd. This keeps the pins straight.
2. Solder the 5 short pin ends on top of the board. Each joint should be a small shiny cone, with no solder bridging two neighbouring pins.

Job 2 · speaker wires on the amplifier (2 joints)

1. Strip about 5 mm of each speaker wire and tin it (melt a little solder into the strands).
2. RED wire through the big hole marked ⊕, BLACK wire through the big hole marked ⊖. Solder, then trim what sticks out. (If the bag has a 2-pin screw terminal, it can go in the two small holes instead, and the wires then screw in.)

Button wire ends, and checks before plugging in



Job 3 · tin the button's wire ends (2 joints, no board)

1. The button already has its two wires. Their bare ends are loose strands that don't grip in the breadboard.
2. Twist each end and melt a little solder into it (tinning), so it becomes stiff like a pin. Or solder each end onto a male jumper pin.

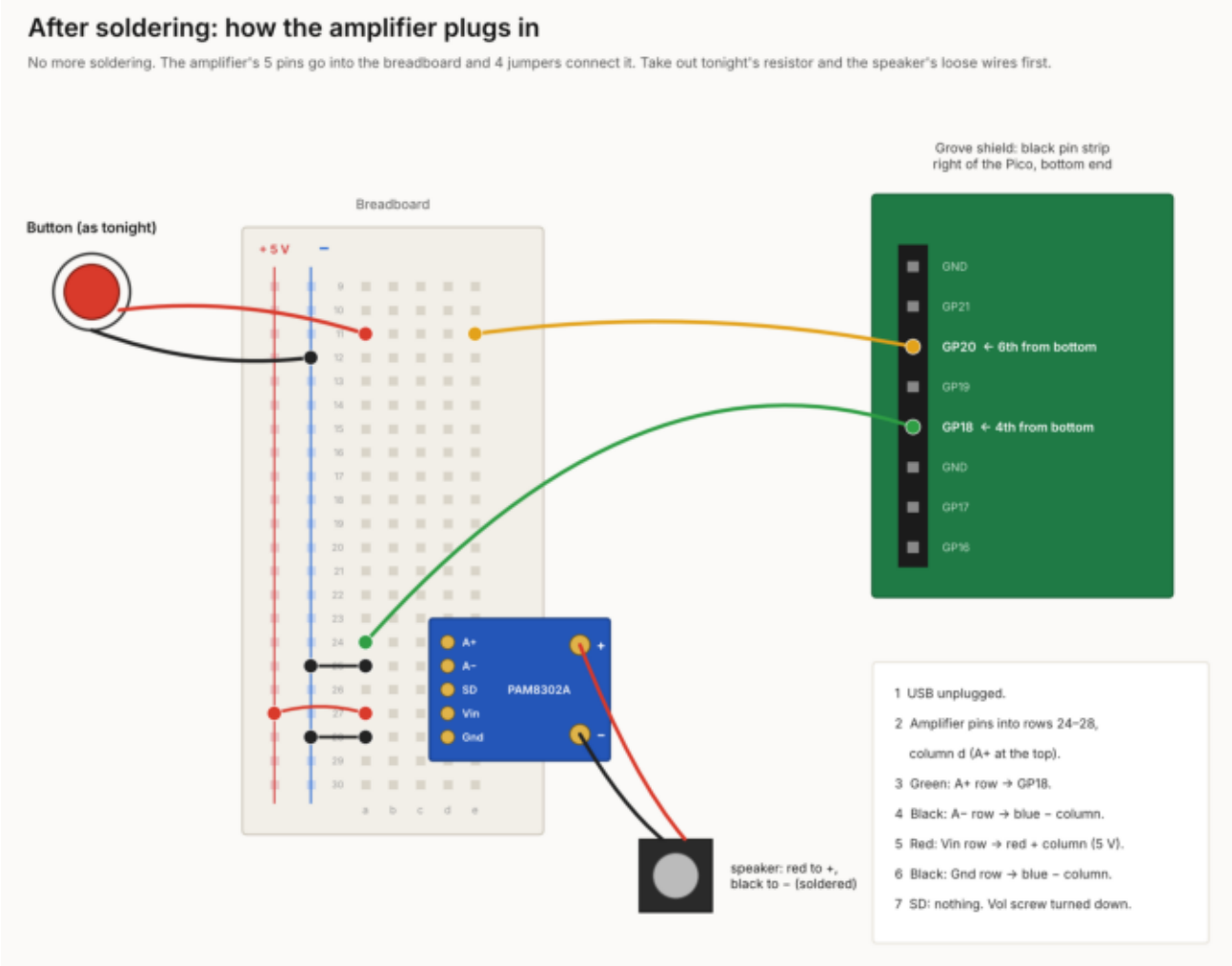
Checks with the multimeter (before anything is plugged in)

1. Continuity mode (the beep): touch Vin and Gnd on the amplifier. It must NOT beep. A beep means a solder bridge: fix it before powering, or it shorts the USB supply.
2. Resistance mode (200 Ω): across the amplifier's ⊕ and ⊖. About 8 Ω means the speaker is connected properly.
3. Look along the 5 header pins: each joint separate and shiny, with no blobs touching the parts next to them.

Why the amplifier: without it, the Pico can only drive the speaker gently through a resistor (a soft beep). The alarm tones (3 pulses, 10 pulses) must be heard across a room.

After soldering: plugging it in (no soldering)

Tonight's resistor and the speaker's loose wires come out; the amplifier goes in their place.



What you should hear

- Two quick rising beeps every time the board starts: the speaker works.
- A beep, and a "BUTTON pressed" line in the live view, each time you press the button.
- Too loud or too quiet: turn the small Vol screw on the amplifier with a small screwdriver. Start low; the USB port also powers the sensor and the light ring.

Pins used (for the report)

- GP18 → amplifier A+ (tone, PWM). GP20 ← button (to GND, internal pull-up). GP16 → light ring (D16 socket). GP0/GP1 ↔ CO2 sensor (UART).
- Amplifier Vin → 5 V; Gnd and A− → GND; SD unconnected (amplifier always on).